

The Balloons Domain

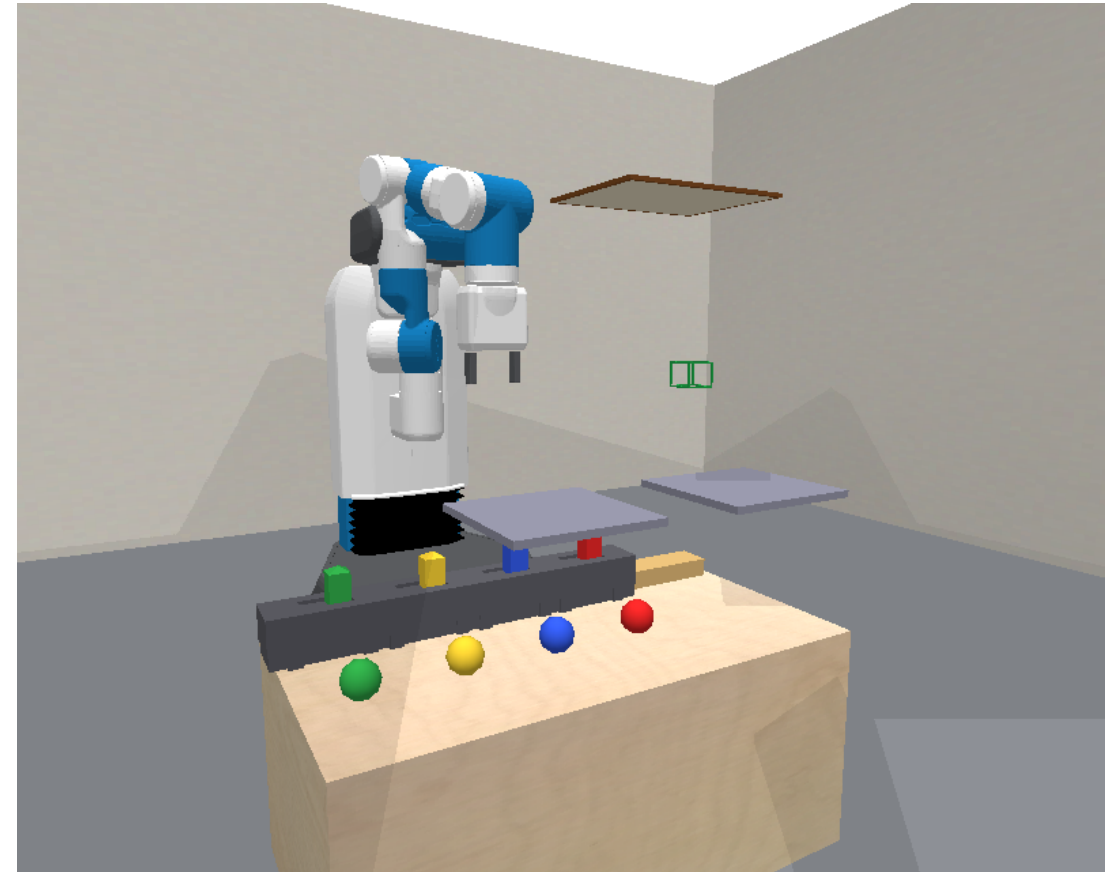
Float a box past obstacles

Choose which balloons to release so the box reaches the green height band at low speed, without bursting a balloon.

The original chute and the new hatch combine hidden lift dynamics with tilt and contact.

Visible attachment ropes, matching clip colours and a ceiling canopy over the payload.

Historical sweep results remain in a separate section.



Current hatch illustration; the ropes appear after release.

What changed from the original puzzle

Transient motion

Drag is **2.2**, down from 12. The rising box can overshoot its predicted rest height.

A balloon can burst at the ceiling even when the predicted hover height is inside the band.

Tilt and contact

Balloon attachments are spread across the box top by clip index.

Off-centre pulls can tilt or sway the box into the **two chute walls**.

Contact challenge

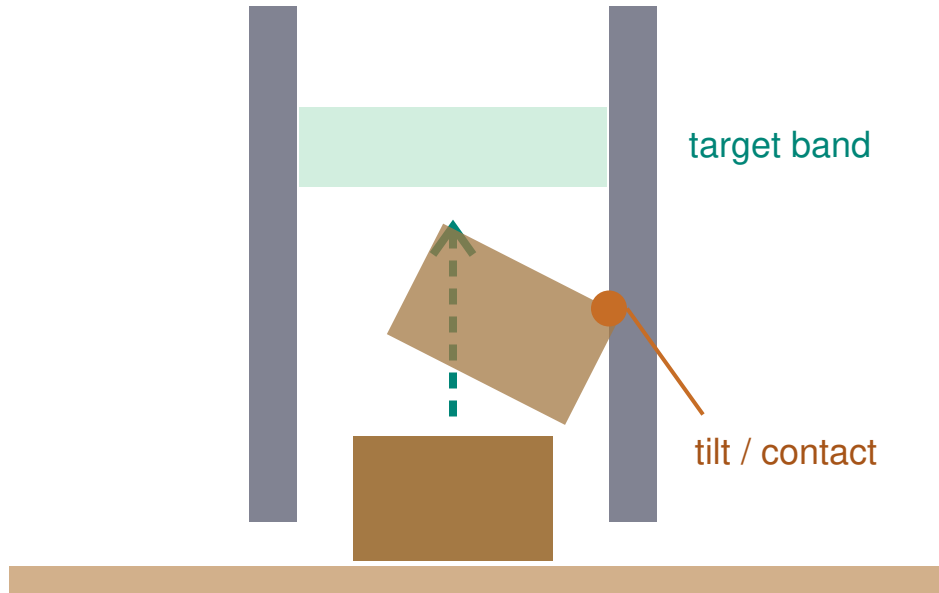
The historical generator searched for alternatives with similar hover heights and different ascent outcomes.

Its early stopping could misclassify an oscillation as a jam. Later audits check sustained contact and full release sequences.

Predicted hover height is a useful filter. It does not certify that the ascent will succeed.

The box must pass between two walls

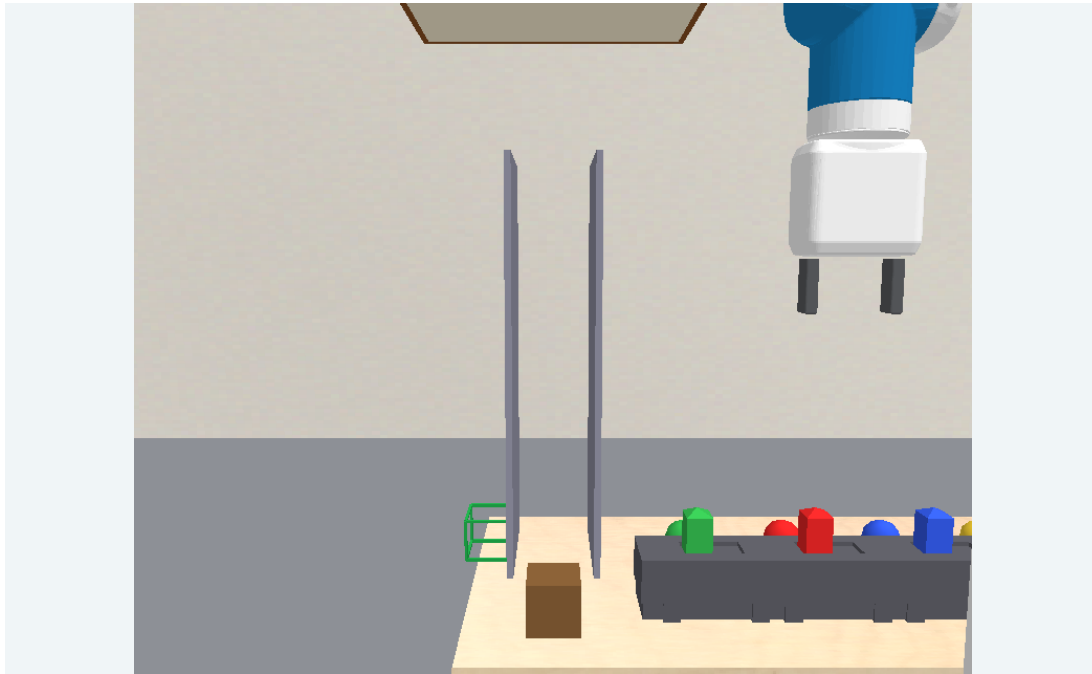
Mechanism schematic, not a recorded trajectory



- The slot is **9.6 cm wide**, from $z = 0.50$ to 1.00 m.
- Attachment offsets span **-2.8 to +2.8 cm**, ordered by clip index.
- Walls collide with **the box only**. The arm and balloons pass through them.
- Balanced net pull alone is not a guarantee: the full motion and contacts determine whether the box clears the slot.

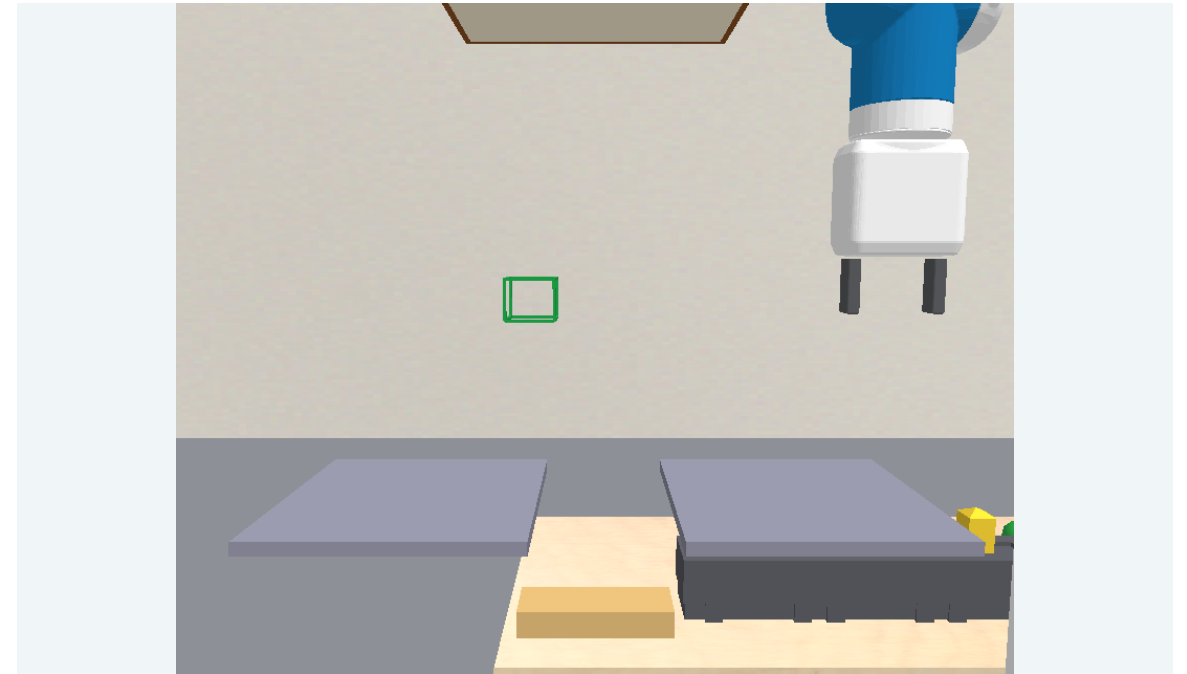
The new hatch changes the path

Original chute



A 7 cm cube fits inside a 9.6 cm vertical slot.
Tilting or swaying can bring the box into a wall.

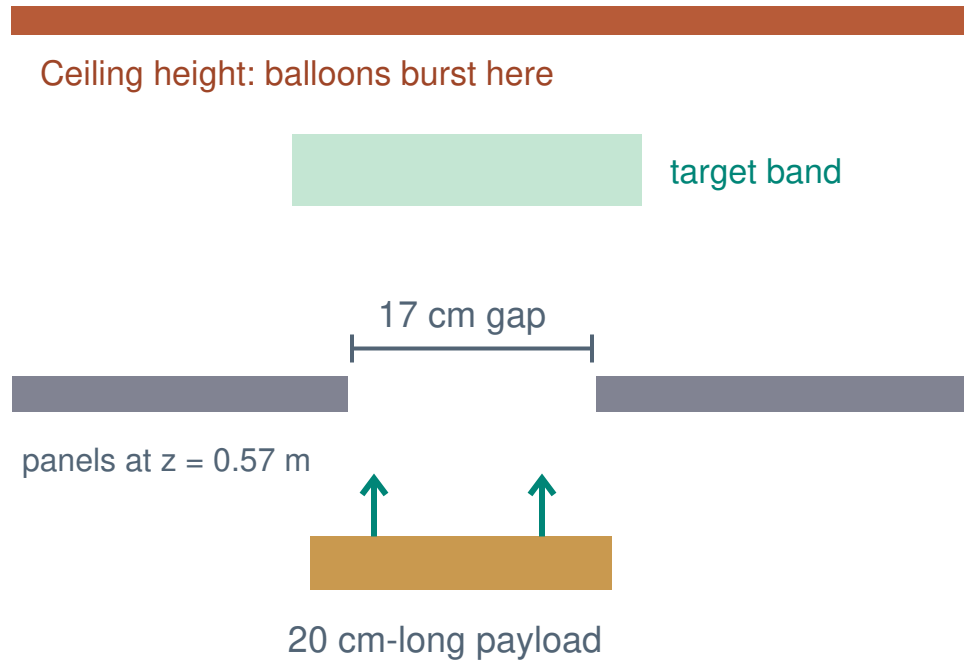
New hatch



A 20 cm-long payload starts below a 17 cm opening.
It must pass the panels and reach the green band above.

Same Release and Wait skills. The hatch makes the box's orientation during ascent central to the task.

Walls block the box; the ceiling bursts balloons

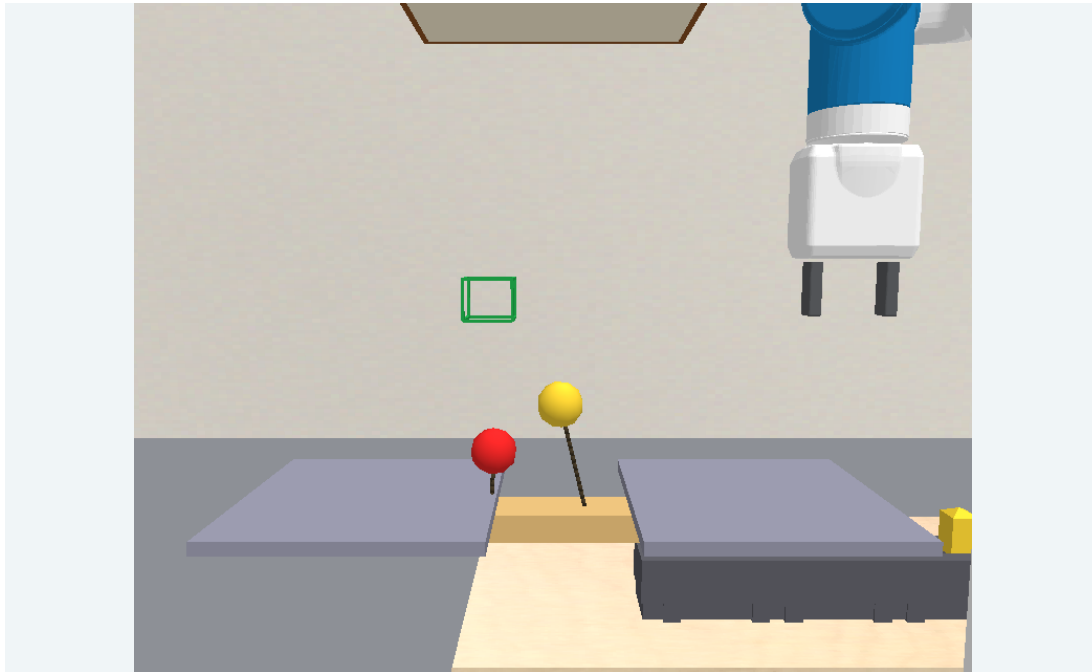


- Both chute walls and hatch panels collide with **the box only**.
- Balloons and the robot pass through these obstacles. Wall contact does **not** burst balloons.
- The hatch opening is offset by 1.2 cm. The payload is $20 \times 7 \times 3.6$ cm.
- Off-centre balloon pulls rotate the box. Release order changes its path and contacts.

Cross-section schematic, not a trajectory. Balloon bursting is a separate ceiling-height rule.

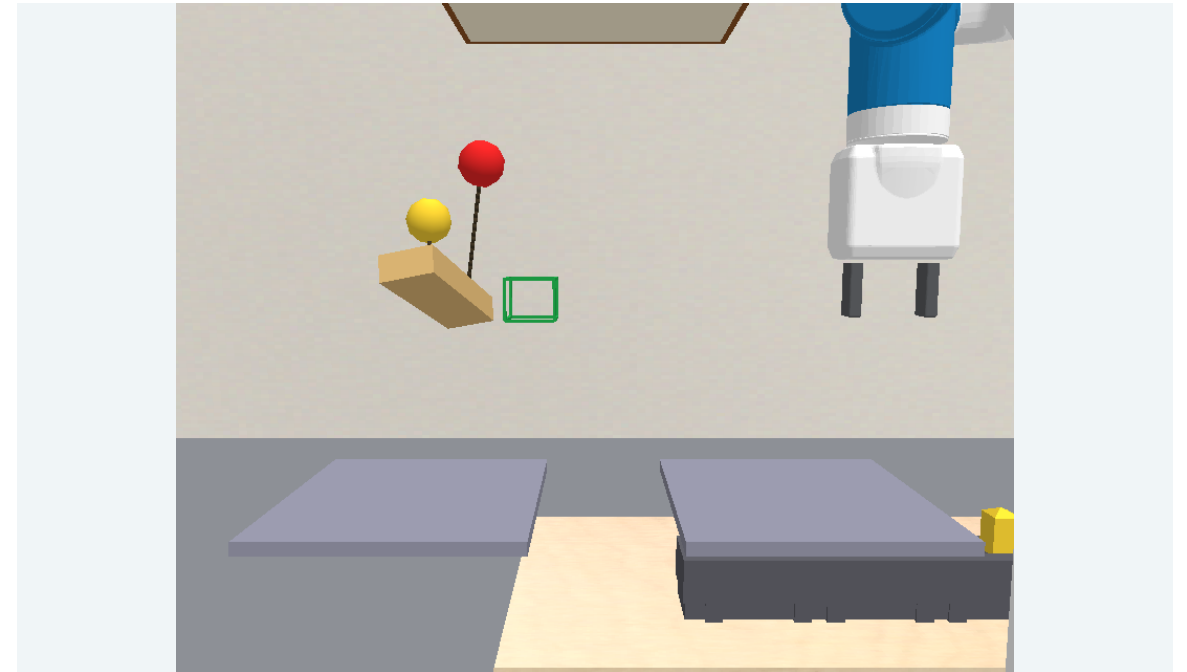
Same two balloons, different release order

Red → gold: jam



65 primitive actions; box centre 54.4 cm.
Persistent support against the hatch; target not reached.

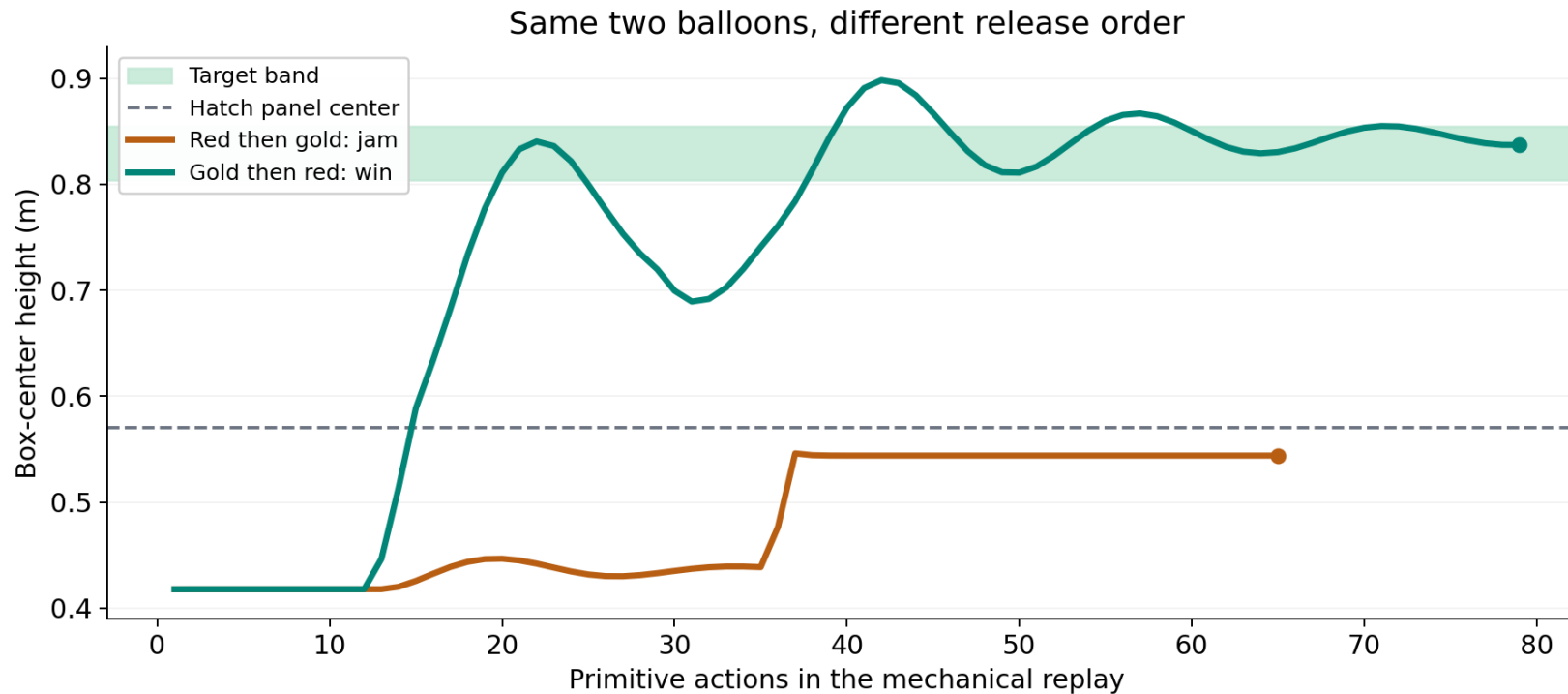
Gold → red: passage



79 primitive actions; box centre 83.7 cm.
The evaluator reports a win inside the target band.

Playback: 16 primitive actions per second, plus start/end holds.
Both clips start from the same task state; no agent chooses these actions.

The difference appears along the trajectory



Verified contact failure. The audited red-then-gold sequence succeeds when box-panel collisions are disabled.

MB hypothesis. A learned simulator could compare release sequences before an irreversible real action. An agent advantage remains unmeasured.

What stays the same, and what must be learned

	Original chute	Hatch prototype
Objective	Hang at rest in a height band	Hang at rest above the hatch
Controls	Release a clip; Wait	The same two skills
Training / test	Two small racks / four-balloon rack	The same compositional split
Hidden physics	Release, colour-dependent lift, mass, drag, burst	The same laws with a wider payload and new contacts
Observed orientation	Absent from the historical box feature vector	Box and balloon roll, pitch and yaw
Pilot noise	Historical sweep below was noiseless	1 cm position; 0.02 rad orientation

Both arms receive the same hatch geometry and noisy observations. MB learns a simulator subclass; MF retains its free-coding baseline.

The clips illustrate physical states from mechanical validation. They are not recordings of noisy agent decisions or evidence of solve-rate improvement.

Same objects and skills, richer motion

Object	Observed features	Role
Box	<code>x, y, z, color, speed</code>	Material identity, position and linear speed; box orientation is absent from this feature vector.
Balloon	<code>x, y, z, color, tied, popped</code>	Gas identity; released/attached status; burst status.
Clip	<code>x, y, z, rot, is_on</code>	Clip <i>i</i> releases balloon <i>i</i> ; an open clip stays open.
Band	<code>x, y, lo, hi</code>	Acceptable box-centre heights, 5 cm apart.
Robot	<code>x, y, z, fingers, roll, tilt, wrist</code>	End-effector and gripper state.

Release(robot, clip)

Push the toggle open. Parameters set approach distance and contact height. A freed balloon cannot be clipped back.

Wait(robot)

Hold still while the physics evolves. A jam is not automatically a terminal loss; it can leave the task unsolved.

Height is only the free-ascent prediction

$$L_c(z) = L_c \cdot \max(0, 1 - (z - z_{table}) / h)$$

$$z^* = z_{table} + h \cdot (1 - W / \Sigma L_c)$$

W includes the box and freed balloons. The second expression assumes an unconstrained equilibrium; it does not model overshoot or wall contact.

Quantity	Sweep value	Meaning
Red / blue / green / gold lift	0.35 / 0.50 / 0.70 / 1.00 N	Per-colour lift at table height
Fade height	0.80 m	Lift declines with height
Pine / oak box mass	0.05 / 0.08 kg	Both materials occur in the sweep
Balloon mass / air drag	0.005 kg / 2.2	Freed mass and transient damping
Ceiling / settle threshold	z = 1.18 m / speed < 0.01 m/s	Burst boundary and the evaluator's low-speed check

A predicted rest height does not prove a jam

1. Draw a rack and box material; enumerate balloon subsets and their unconstrained hover heights.
2. The sweep's helper could stop at a low-speed turning point before the box finished oscillating.
3. Later replay showed that some supposed jamming alternatives actually reached the target. The earlier uniqueness and jam labels are not established.
4. The revised checks execute Release sequences, require sustained off-goal rest, and verify that a suspected jam disappears without wall collisions.

Height alone can favour the wrong subset. Release order and the path through the chute still matter during execution.

Timeouts remain unresolved. A tested reference can win in both orders while an alternative wins in one order and jams in the other.

Two near-identical height predictions

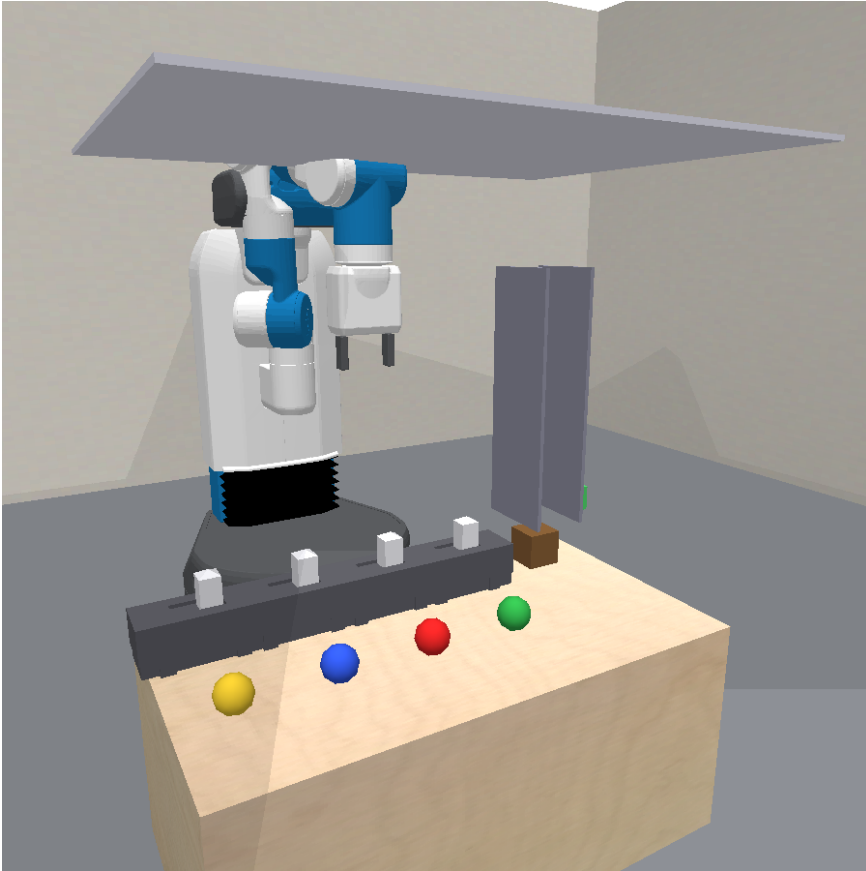
Oak box; rack order: green, red, blue, gold.
Recorded target band: **[0.5023, 0.5523] m**.

Subset	Total lift	Analytic z*	Interpretation
Gold (clip 3)	1.00 N	0.5329 m	In band; the recorded agent released it and won.
Green + red (clips 0, 1)	1.05 N	0.5273 m	Also in band, closer to its centre; height cannot certify it.

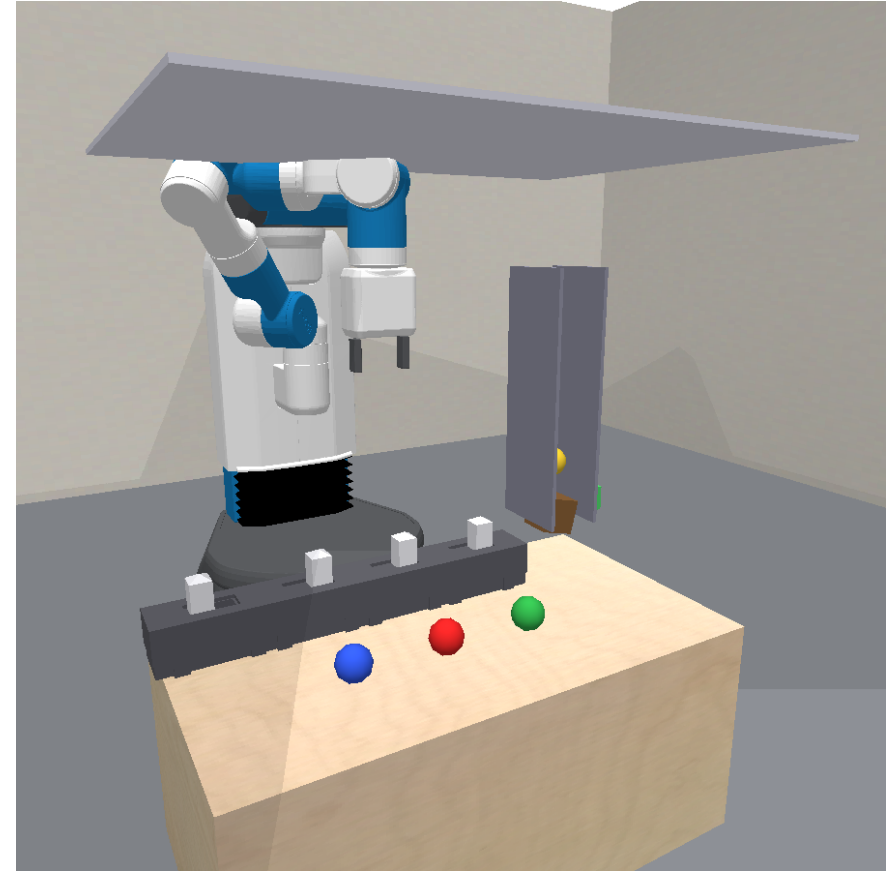
The predictions differ by only **5.6 mm**. The old isolated-hover-height example no longer describes this task.

The successful real sequence was Release(clip3), 41 steps, then Wait, 7 steps. The alternative's height above is calculated from the source law, not a newly replayed counterfactual.

Gold alone solves this test level



Before: four clipped balloons, box below the chute.



After: gold released; the evaluator reports WIN at test step 48.

The chute partially hides the box and green band in this camera view. These are preserved experiment frames, not a replay on newer code.

Three levels per seed, one continuing agent

	Training	Test
Levels per seed	2	1
Balloons per level	2 or 3	4, the full palette
Box material	Pine or oak	Pine or oak
Resets	Allowed and counted	Unavailable

- Training draws prefer colours and materials not yet shown; generation can fall back to a free draw.
- The journal, learned model and recorded data carry forward to each new level. Test composes familiar identities in a new rack.
- The sweep uses seeds 0–2 and noiseless observations. Report solve rate, total real steps and agent resets over **all train + test levels**.

What the simulator adds

Model-based (MB)

- Uses a visible base simulator with the scene, chute geometry and rigid-body contacts.
- Writes a simulator subclass for release, attachment, lift and bursting; explicitly fits physical parameters.
- Can test release plans in its learned simulator before paying real steps.

Model-free baseline (MF)

- Has the same real environment, skill tools, observations, journal and recorded experience.
- Has no supplied simulator or fit-and-rollout workbench.
- Still has arbitrary Python: it can fit equations and reason about dynamics by hand.

Contact prediction is the intended benefit. The baseline's name does not mean it must guess or cannot build an analytic model.

Balloons: a modest gap in these three seeds

Arm	Seed	Levels solved	Total real steps	Resets
MB	0	3 / 3	273	0
MB	1	3 / 3	385	1
MB	2	3 / 3	787	1
MB	Mean	100.0%	481.7 (n=3)	0.67
MF	0	3 / 3	506	1
MF	1	2 / 3	558	1
MF	2	3 / 3	423	1
MF	Mean	88.9%	464.5 (n=2)	1.00

MB: 9/9 levels; MF: 8/9. These small samples do not establish a large step-efficiency advantage.

Mean steps include only whole-run successful seeds, including their attempts and resets (qualifying n shown). Solve rate and mean resets include all seeds. Per-seed rows retain actual total steps.

Original results and hatch evidence have separate sources

Source	What to inspect
<code>predicators/envs/pybullet_balloons_base.py</code>	Scene, chute collision filters, attachment offsets and observed features.
<code>predicators/envs/pybullet_balloons.py</code>	Lift, attachment, burst, task generation, oracle checks and evaluator.
<code>predicators/settings.py</code>	Drag 2.2; contact-decoy default; 2 cm tolerance; training palette.
<code>protocol_continual_uniform_sweep.yaml</code>	Two training levels, common MB/MF flags and noiseless protocol.
<code>docs/envs/balloons/sweep_20260907/</code>	Saved screenshots, source manifest, per-seed results and hashes.
<code>docs/envs/balloons/visuals/</code>	Hatch videos, frame images, height curves and a replay manifest.
<code>docs/slides/make_balloons_domain_slides.py</code>	Rebuilds both standalone HTML decks; <code>--pdf</code> also renders the PDF.

Hatch videos replay an audited task with fixed release sequences. Physical renders show true poses; the observation-noise channel is not painted into the scene. No hatch MB/MF result is claimed.